CAN OGAN KARAGUN

Software Engineer

cankaragun@gmail.com | coganka.com | github.com/coganka | linkedin.com/in/canogankaragun

Software engineer building scalable, data-intensive systems for telecom operations. Experience applying algorithms and data structures to performance optimization, network analysis, and distributed services. Shipped games on Xbox and iOS.

EXPERIENCE

Software Engineer - TechNarts, Ankara, Turkey | *Sep 2025 - Present*

STAR Suite - Telecom Operations Support Platform (OSS) for Turkcell

- Selected from the existing STAR Suite team to join a 3-engineer platform modernization team in Feb 2026, rebuilding the system with Django, PostgreSQL, Redis, Kafka, and React.
- Reduced production-critical page response times from 20-30 seconds to 200-400 ms by eliminating N+1 queries, introducing snapshot-based read models, and adding Redis caching.
- Migrated telecom datasets containing hundreds of thousands to millions of records from MariaDB to redesigned PostgreSQL schemas using SQL migration pipelines.
- Developing graph-based fiber routing and network impact analysis to identify downstream effects of faults across telecom topologies.
- Consolidated Huawei, Nokia, and ZTE-specific logic into reusable backend services; built Huawei 400G inventory management covering equipment lifecycles and topology dependencies.
- Integrated vendor network management systems through REST, SOAP, and SSH-tunneled connections to synchronize inventory and performance data.

Software Engineer Intern - KordSA | *2025*

- Developed Python tools for data processing and visualization to support analysis workflows.

Software Engineer Intern - Kocaeli University IT Department | *2023*

- Implemented Redis caching for internal services.

SELECTED PROJECTS

- **API Gateway** - Built a Flask reverse proxy with PostgreSQL-backed API keys, Redis rate limiting, retries, circuit breaking, and Prometheus metrics.
- **Redis Job Queue** - Built concurrent Go workers with Redis Streams, scheduled jobs, exponential-backoff retries, job status tracking, and a dead-letter queue; packaged with Docker Compose.
- **Go Distributed Microservices** - Built distributed backend services using Go, RabbitMQ, gRPC, and Docker.
- **3D Graphics Library / Rasterizer** - Built a software renderer from scratch in C, with matrix transformations, clipping, lighting, and textured triangle rasterization.
- **2D Physics Engine** - Built a physics engine in C++ with rigid-body simulation, collision handling, and constraints.
- **SLAM Mapping Simulation** - Built a robot simulation to map unknown environments using laser scans and Kalman filtering.
- **Real-Time Hand Tracking Game** - Built a Unity game controlled by real-time hand tracking using Python and OpenCV.

PUBLISHED GAMES

Xbox: Danger Close and Fight Freaks (2021) - **70K+ combined downloads**.
iOS: Encounter and Zigzag Chicken (2022) - **35K+ combined downloads**.

TECHNICAL SKILLS

Languages: Python, Go, SQL, JavaScript, C, C++, C#
Frameworks & Data: Django, Flask, Node.js, PostgreSQL, Redis, Kafka, RabbitMQ, REST APIs, gRPC
Tools & Platforms: Git, Linux, Docker, GitHub Actions, Unity, OpenCV

EDUCATION

Kocaeli University | B.Sc. Electronics & Telecommunications Engineering | May 2025